# T3 A_-4: general EFT1 RGEs

Matchete/RGBeta symbolic output in mathematical notation

## A_-4

Additional quartics: 0, additional bilinears: 0, extra Yukawa: none.
*General matching and Wilson coefficients are not computed in this symbolic report.*

## Coupling and interaction notation

| symbol | interaction / definition |
|---|---|
| $g_1$ | $D_\mu \supset i g_1 Y B_\mu$ |
| $g_2$ | $D_\mu \supset i g_2 T^A W_\mu^A$ |
| $g_3$ | $D_\mu \supset i g_3 t^A G_\mu^A$ |
| $Y_u$ | $-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$ |
| $Y_d$ | $-\bar{Q} Y_d H d + \text{h.c.}$ |
| $Y_e$ | $-\bar{L} Y_e H e + \text{h.c.}$ |
| $y_1$ | $L F S_1 + \text{h.c.}$ |
| $y_2$ | $L F S_2 + \text{h.c.}$ |
| $M_F$ | $-\bar{F} M_F F$ |
| $T$ | $\text{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$ |
| $T_\nu^{(1)}$ | $\text{Tr}(y_1^\dagger y_1)$ |
| $T_\nu^{(2)}$ | $\text{Tr}(y_2^\dagger y_2)$ |
| $T_\nu$ | $T_\nu^{(1)} + T_\nu^{(2)}$ |
| $m_1^2$ | $m_1^2 S_1^\dagger S_1$ |
| $m_2^2$ | $m_2^2 S_2^\dagger S_2$ |
| $\lambda_1$ | $\frac{1}{2}\lambda_1 (H^\dagger H)^2$ |
| $\lambda_{S_1}^{(1)}$ | $\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$ |
| $\lambda_{S_2}^{(1)}$ | $\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$ |
| $\lambda_{HS_1}^{(1)}$ | $\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$ |
| $\lambda_{HS_2}^{(1)}$ | $\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$ |
| $\lambda_{HS_1}^{(A)}$ | $\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$ |
| $\lambda_{HS_2}^{(A)}$ | $\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$ |
| $\lambda_5$ | $\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$ |
| $\lambda_{12}^{(1)}$ | $\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$ |
| $\lambda_{12}^{(A)}$ | $\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$ |
| $\lambda_{12}^{(\times)}$ | $\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$ |
| $\lambda_{S_1}^{(A)}$ | $\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$ |
| $\lambda_{S_2}^{(A)}$ | $\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$ |
| $\alpha$ | $Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$ |
| $C_2(d)$ | $(d^2-1)/4$ |
| $T(d)$ | $d(d^2-1)/12$ |
| $C_{LLS_1S_2}^{ij;AB}$ | $L_i L_j S_1^A S_2^B$ |
| $C_{ijab}$ | $L_i L_j \phi_a \phi_b$ |
| $C_5^{ij}$ | $(L_i^T C L_j) H H$ |
| $R_W$ | $16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$ |

## General-interaction coupling index

| Coupling | Field content / definition | Contraction / source |
|---|---|---|

Each $X_i$ labels one field-content class in the general-quartic registry; a superscript distinguishes different invariant contractions where present. The original RGBeta identifier and contraction are retained in the index. This is notation only, not a physical normalisation mapping.

## One-loop renormalisable beta functions

$16\pi^2\, dX/d\ln\mu = \beta_X$

$g_1$

$$16\pi^2 \frac{dg_1}{d\ln\mu} = \frac{55{g_1}^3}{6}$$

$g_2$

$$16\pi^2 \frac{dg_2}{d\ln\mu} = -\frac{5{g_2}^3}{2}$$

$g_3$

$$16\pi^2 \frac{dg_3}{d\ln\mu} = -7{g_3}^3$$

$Y_u$

$$\begin{aligned}
16\pi^2 \frac{dY_u}{d\ln\mu} = {} & 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_u)_{ij} \\
& + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_u)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_u)_{ij} \\
& - \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_u\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_u\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_u)_{ij} \\
& - 8{g_3}^2\,(Y_u)_{ij} \\
& - \frac{17}{12}{g_1}^2\,(Y_u)_{ij}
\end{aligned}$$

$Y_d$

$$\begin{aligned}
16\pi^2 \frac{dY_d}{d\ln\mu} = {} & \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij} \\
& + \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_d\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_d\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_d)_{ij} \\
& - 8{g_3}^2\,(Y_d)_{ij} \\
& - \frac{5}{12}{g_1}^2\,(Y_d)_{ij}
\end{aligned}$$

$Y_e$

$$\begin{aligned}16\pi^2 \frac{dY_e}{d\ln\mu} = &3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_e)_{ij}\\ &+3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_e)_{ij}\\ &+\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_e)_{ij}\\ &+\frac{3}{2}\Big(Y_e\ (Y_e)^\dagger\ Y_e\Big)_{ij}\\ &-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}\\ &-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}\end{aligned}$$

$m_1^2$

$$\begin{aligned}16\pi^2 \frac{dm_1^2}{d\ln\mu} = &-24{g_1}^2 m_1^2\\ &+6\lambda_{12}^{(1)} m_2^2\\ &+4\lambda_{S_1}^{(1)} m_1^2\end{aligned}$$

$m_2^2$

$$\begin{aligned}16\pi^2 \frac{dm_2^2}{d\ln\mu} = &-12{g_2}^2 m_2^2\\ &-6{g_1}^2 m_2^2\\ &+2\lambda_{12}^{(1)} m_1^2\\ &+m_2^2(8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})\end{aligned}$$

$\lambda_1$

$$\begin{aligned}16\pi^2 \frac{d\lambda_1}{d\ln\mu} = &\lambda_1(12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))\\ &-12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big)\\ &-4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big)\\ &-12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)\\ &+4\lambda_5\,(\lambda_5)^*\\ &+\frac{9{g_2}^4}{4}\\ &+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)\\ &-9{g_2}^2\lambda_1\\ &+\frac{3{g_1}^4}{4}\\ &+12{\lambda_1}^2\\ &+2{\lambda_{HS_1}^{(1)}}^2\end{aligned}$$

$$
\begin{aligned}
&+ 6{\lambda_{HS_2}^{(1)}}^2 \\
&+ 3{\lambda_{HS_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{S_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_1}^{(1)}}{d\ln\mu} = &192{g_1}^4 \\
&- 48{g_1}^2\lambda_{S_1}^{(1)} \\
&+ 6{\lambda_{12}^{(1)}}^2 \\
&+ 4{\lambda_{HS_1}^{(1)}}^2 \\
&+ 10{\lambda_{S_1}^{(1)}}^2
\end{aligned}
$$

$\lambda_{S_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(1)}}{d\ln\mu} = &22{g_2}^4 \\
&+ {g_1}^2\left(8{g_2}^2 - 12\lambda_{S_2}^{(1)}\right) \\
&- 24{g_2}^2\lambda_{S_2}^{(1)} \\
&+ 12{g_1}^4 \\
&+ 2{\lambda_{12}^{(1)}}^2 \\
&+ 4{\lambda_{HS_2}^{(1)}}^2 \\
&+ 14{\lambda_{S_2}^{(1)}}^2 \\
&+ 8\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\
&+ 16{\lambda_{S_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{HS_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_1}^{(1)}}{d\ln\mu} = &\lambda_{HS_1}^{(1)}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)) \\
&+ 12\lambda_5\,(\lambda_5)^* \\
&- \frac{9{g_2}^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 12{g_1}^4 \\
&- \frac{51{g_1}^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_1}^{(1)} \\
&+ 4{\lambda_{HS_1}^{(1)}}^2 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}
\end{aligned}
$$

$\lambda_{HS_2}^{(1)}$

$$\begin{aligned}16\pi^2 \frac{d\lambda_{HS_2}^{(1)}}{d\ln\mu} = &\lambda_{HS_2}^{(1)}(6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 4\lambda_{S_2}^{(A)})\\ &+ 4\lambda_5\,(\lambda_5)^*\\ &+ 6{g_2}^4\\ &- \frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2}\\ &+ 3{g_1}^4\\ &- \frac{15{g_1}^2\lambda_{HS_2}^{(1)}}{2}\\ &+ 2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}\\ &+ 6\lambda_1\lambda_{HS_2}^{(1)}\\ &+ 4{\lambda_{HS_2}^{(1)}}^2\\ &+ 8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}\\ &+ 6{\lambda_{HS_2}^{(A)}}^2\end{aligned}$$

$\lambda_{12}^{(1)}$

$$\begin{aligned}16\pi^2 \frac{d\lambda_{12}^{(1)}}{d\ln\mu} = &4\lambda_5\,(\lambda_5)^*\\ &- 12{g_2}^2\lambda_{12}^{(1)}\\ &+ 48{g_1}^4\\ &- 30{g_1}^2\lambda_{12}^{(1)}\\ &+ 4{\lambda_{12}^{(1)}}^2\\ &+ 4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}\\ &+ 8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}\\ &+ 4\lambda_{12}^{(1)}\lambda_{S_2}^{(A)}\\ &+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}\end{aligned}$$

$\lambda_5$

$$\begin{aligned}16\pi^2 \frac{d\lambda_5}{d\ln\mu} = &\lambda_5(6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 4\lambda_{HS_2}^{(A)})\\ &- \frac{21{g_2}^2\lambda_5}{2}\\ &- \frac{33{g_1}^2\lambda_5}{2}\\ &+ 2\lambda_{12}^{(1)}\lambda_5\\ &+ 2\lambda_1\lambda_5\\ &+ 4\lambda_{HS_1}^{(1)}\lambda_5\\ &+ 4\lambda_{HS_2}^{(1)}\lambda_5\end{aligned}$$

$\lambda_{HS_2}^{(A)}$

$$\begin{aligned}16\pi^2 \frac{d\lambda_{HS_2}^{(A)}}{d\ln\mu} = \lambda_{HS_2}^{(A)}(&6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 16\lambda_{S_2}^{(A)}) \\ &+ \frac{8}{3}\lambda_5\,(\lambda_5)^* \\ &+ {g_1}^2\left(-4{g_2}^2 - \frac{15\lambda_{HS_2}^{(A)}}{2}\right) \\ &- \frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2} \\ &+ 2\lambda_1\lambda_{HS_2}^{(A)} \\ &+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)} \\ &+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}\end{aligned}$$

$\lambda_{S_2}^{(A)}$

$$\begin{aligned}16\pi^2 \frac{d\lambda_{S_2}^{(A)}}{d\ln\mu} = &-2{g_2}^4 \\ &+ {g_1}^2\left(8{g_2}^2 - 12\lambda_{S_2}^{(A)}\right) \\ &- 24{g_2}^2\lambda_{S_2}^{(A)} \\ &+ 3{\lambda_{HS_2}^{(A)}}^2 \\ &+ 12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\ &+ 20{\lambda_{S_2}^{(A)}}^2\end{aligned}$$